



MANTA(ray): A Multi-Sensor Autonomous Navigation & Transit Architecture

GPS-IMU Sensor Fusion RoboCar with Object Identification for Obstacle Avoidance

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Proposal Details and Progress

Must-Haves:

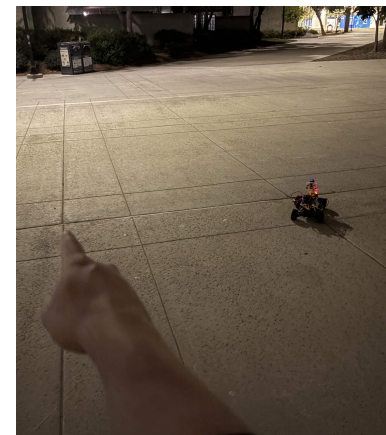
- GPS-IMU Sensor Fusion ✓
- Autonomous Point-to-Point Navigation ✓ (*achieved semi-autonomous*)
- OAK-D Lite YOLO Object Identification ✓
- OAK-D Lite depth calibration ✓
- Obstacle avoidance software ✓

Reach Goals:

- LiDAR Detection Emergency Stop (*incorporated into obstacle detection stack w/OAK-D*)
- Multiple selectable pickup and destination locations.
- A simple taxi-request interface for selecting the desired destination.
- Automatic continuation after an obstacle is removed. ✓
- Vehicle-speed-dependent LiDAR stopping distances.
- Improved route following using waypoint tracking or a path-following controller.
- Dynamic obstacle avoidance rather than emergency stopping only.
- Visualization or logging of the estimated GPS, IMU, and fused vehicle trajectories

What Went Right:

- OAK-D produced normal cone bounding
- GPS-IMU Sensor Fusion
 - Able to maintain constant rate of 100 hz
 - 2-4m accuracy
- Navigation stack able to command movement through safety gate
- IMU, LD06 integrated successfully

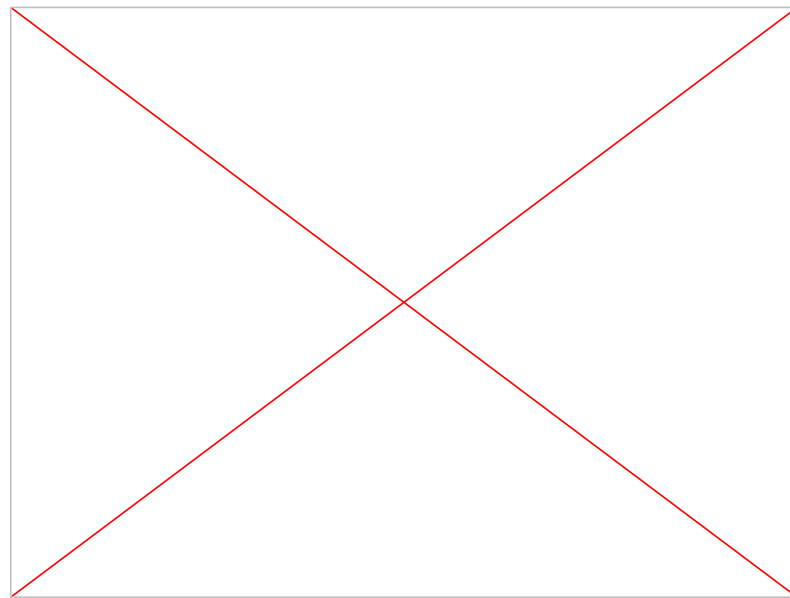


Partial and Full Integration:

Partial:



Full:



What Did Not Work

Observed problems:

- Two-cone lane-guidance handoff was not reliable under live motion
- Camera wrapper could keep the OAK-D device locked after restart
- LiDAR health flickered under integration load
- Multiple navigation errors resulted in stressful environment
- Difficulty with component stabilization



Cause of Issues

Individual components worked (for the most part), but the integrated stack depended on handoffs between perception, avoidance, waypoint navigation, and the safety gate → duplicated publishers pushing robot into stop or unstable steering when doing live tests.



How We Would Solve the Issues

Software Architecture

- Create single explicit state machine: GPS → gate approach → visual servo → clearance → GPS
- Guarantee exactly one publisher for waypoint navigation
- Add watchdogs that report stale topics without changing state

Testing Fixes

- Run perception-only tests before any motion testing
- Use rosbag replay for repeatable cone scenarios
- Tune steering sign and gain w/wheels lifted before floor tests

If We Had Another Week

- Start and finish of beautification of the robot / hardware —>
- Extensive debugging of ros2 nodes integration for full-stack
- Increase LiDAR and OAK-D stabilization / reduce flimsiness



Gantt Chart

3 main tasks: GPS-IMU Sensor Fusion & Localization, YOLO Obstacle Detection Model, Taxi System Integration & Validation

